

Instructor: Date:

Learner: Regional Branch:

Program: Center:

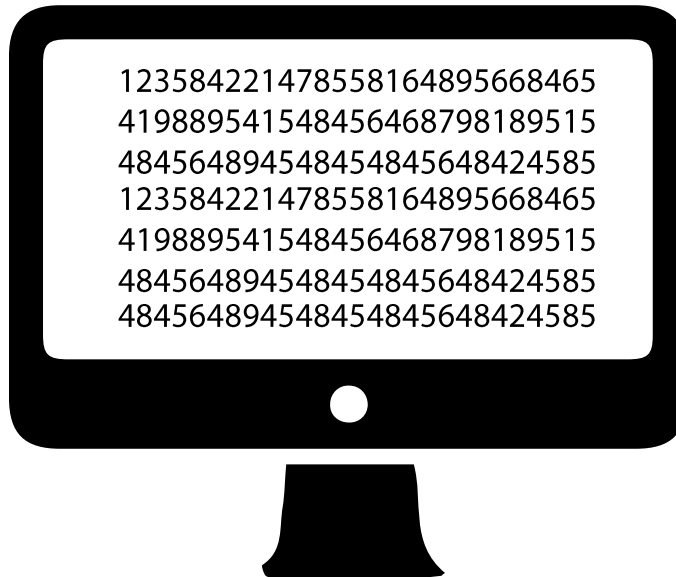
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Let's Start!

This is how my computer works.



Learning Outcome:

Identify information about algorithms related to software programming in simple spoken and written situations.

Ask your partner:

- Do you use your computer every day? Why?
- What kind of information do you usually search on Internet?

Can Do Statements:

- I can talk about different tasks a computer does.
- I can say what an algorithm is.
- I can discuss how an algorithm works.





Let's Explore!

A teacher and her student are talking about what an algorithm is.



Instructions: Read the text below.

Mr. West: Hi Monique, nice to see you!

Monique: Good morning Mr. West, nice to see you too.

Mr. West: Tell me, why do you need a tutoring session?

Monique: Because I need help with the algorithm topic, there are some things that I don't understand.

Mr. West: Ok, I see.

Monique: In class, you always say that an algorithm is a procedure which permits a computer to solve a problem, but last class you said that it is a series of unambiguous instructions, what does it mean?

Mr. West: Well, it means that there is no any kind of subjective interpretation of the algorithm. A computer performs all kinds of operations you want. For example, it makes a spell check when you finish a text, or it uses a search engine when you need to seek information. Therefore, computers need the algorithms to do all these procedures. Tell me, what does it mean to you?

Monique: I understand that we use an algorithm to make a computer solve a problem. Every time I command the computer to do a task, it uses the same algorithm to solve the problem, it always works like this, and there are no other different interpretations of it.

Mr. West: Excellent! You are a quick learner, but how does the computer understand the instructions that are part of the algorithm?

Monique: That's easy! You need to use a programming language so that the computer can understand the algorithm. Does the algorithm always work like that?

Mr. West: Not really, there are now different ways to optimize the process. These approaches make the algorithm solve the problem faster and efficiently.

Monique: Does it work algorithm by algorithm?

Mr. West: Yes, it does. You optimize the algorithm to solve a particular problem at a time.





Let's Practice!

Let's say what an algorithm is.

Instructions: Read the sentences. Write the missing word(s) in the blank. Use: those instructions, this procedure, That means.

An algorithm is as simple as a procedure. _____ permits a computer to solve any problem. It is also a series of unambiguous instructions. _____, _____ can't be given with subjective interpretations.



Let's Practice!

Let's answer these questions related to algorithms!

Instructions: Use the words from the box to correctly answer the questions.

Word Bank

I need to clarify the algorithm topic. / In different ways, to optimize the process. / Yes, a computer needs algorithms. / I think it is useful for a computer to solve a problem. / There can't be any subjective interpretation of the algorithm.

1. Why do you need a tutoring?

2. What does unambiguous instructions mean?

3. Does a computer need the algorithms to make certain tasks?

4. What does algorithm mean to you?

5. How does an algorithm work?



Extension Activity

Let's see if you have any questions!

Instructions: Complete the following steps. Fill in the Self-Assessment.

- Search for information about algorithms.
- Identify things you find difficult to understand about algorithms.
- Make a list of questions you have about algorithms.
- Share your questions and discuss about them with your teacher.

Self-Assessment

Yes

No

- I search for information about algorithms.
- I identify things I find difficult to understand about algorithms.
- I make a list of questions I have about algorithms.
- I share information and discuss about the topic.

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Target Words

- unambiguous** (adjective): something that is unclear
e.g. but last class you said that it is a series of unambiguous instructions, what does it mean?
- subjective** (adjective): something that can have different interpretations
e.g. Well, it means that there is no any kind of subjective interpretation of the algorithm.
- search engine** (noun): a computer program that searches for information on the web worldwide
e.g. It makes a spell check when you finish a text, or it uses a search engine when you need to seek information.
- optimize** (verb): to make something better, make it best
e.g. Not really, there are now different ways to optimize the process.
- task** (noun): a specific piece of work assigned to a person or computer
e.g. Every time I present a task to the computer, it uses the same algorithm to solve the problem.
- interpretation** (noun): a conception or idea about something
e.g. It always works like this, and there are no other different interpretations of it.



Learning Points

Simple Present Interrogative Form

You use the **present simple tense** to talk about facts and events that happen in your daily life like routines or habits.

You use the **simple present tense** in the interrogative form to ask questions about things that happen frequently.

Remember that it's important to use the auxiliary verb **"do"** or **"does"** before the subject.

Let's see the structure:

Auxiliary verb do/ does + subject + main verb + complement + ?

Example:

Does he understand algorithms?

Note: In the interrogative form, contrary to the affirmative form, the auxiliary verb changes, and the main verb remains in the base form.

Example:

He understands algorithms.

Does he understand algorithms?

You can also use Wh- questions to get the information in detail. Let's see the following structure:

Wh-question + **auxiliary verb do/ does** + subject + main verb + complement + ?

Example:

Where do you study programming?

Examples

- So tell me, why do you need a tutoring session?
- Does the algorithm always work like that?





Learning Points

Algorithms Alternative Approaches and Optimization

The Algorithms can solve problems on a computer using a specific procedure.

It is necessary to have a procedure (called algorithm), with specific steps that lead to an answer for each computer problem. Some procedures can include looking for specific information on a large source of information. When the algorithm looks word by word for the information, it can take a very long time to obtain an answer because hundreds of algorithms appear at the same time.

Some alternative approaches help to optimize the algorithms. For example, the algorithm optimization can find a solution in a fast way, so we can say that one of the most important aspects to improve is the time.